Final System Architecture: Baseline vs. Our Proposed Robust Defense (V9)

CSE-CIC-IDS2018 Benchmark | SHAP Adversarial-Stability Feature Selection | Pure-GPU Online PGD Training

STAGE 1 & 2: DATA INGESTION, PREPROCESSING, LEAKAGE-FREE PARTITIONING & SMOTE BALANCING

1. Raw Flow Ingestion

CSE-CIC-IDS2018 (10 CSVs)
8,284,871 Total Flows
14 Attack Types + Benign

2. Data Sanitization

Drop IDs, DstPort
Drop Infs/NaNs & Constants
6,320,003 Clean Flows

3. Stratified Partitioning

80% Train (4,550,401)
20% Test (1,264,001 Isolated)
10% Dev Split (505,601)

4. Scaling & MI Filter

StandardScaler (Train only)
Mutual Information (MI)
Top-40 Candidate Features

5. SMOTE Balancing

Applied to Core-Train
7,191,590 Balanced Rows
(1:1 Equalized Ratio)

STAGE 3 & 4: FEATURE SPACE SELECTION & ADVERSARIAL MODEL TRAINING

Model 1: Baseline MLP

Feature Space: Standard (15 features)
Training: Clean Balanced Data (7.19M rows)
50 Epochs, Adam (lr=0.001)
Batch Size: 2048

Model 2: Our Proposed Defense (V9)

Standard Space | V9 Robust Space

Feature Space: V9 Robust Space (27 features)
Phase 1: 10 ep Clean Pretraining (Warmup)
Phase 2: 40 ep Pure-GPU Dynamic Online PGD
Real-Time Per-Batch Attack Injection (k=5, eps=0.05)

SHAP Adversarial-Stability Selection [NOVELTY]

Calibrate proxy model on training pool
Compute SHAP(clean) & SHAP(adv)
S(f) = 1 - (|clean - adv| / [clean + adv + eps])
• Standard Feature Space: Top-15 Standard Features
• V9 Robust Feature Space: Top-27 Features (80% Stability Weight)

STAGE 5: COMPREHENSIVE MULTI-THREAT EVALUATION SUITE (SEEDS: 42, 123, 2024)

A. Standard & White-Box

• Clean Test Traffic (1.26M flows)
• FGSM Attack (eps=0.05)
• PGD-100 Attack (100 steps, eps=0.05)
V9 Gain: +3.37% Clean Acc
70.0% PGD ASR Reduction (6.05% ASR)

B. Unseen Cross-Attacks

• Auto-PGD (APGD-CE Adaptive)
• MIM (MI-FGSM Momentum, mu=1.0)
• BIM (I-FGSM Clean-start)
• CW-L_inf Margin Loss Attack
V9 Gain: 53.5% - 73.8% ASR Reduction

C. Black-Box & Probes

• Square-style Score Search
• Transfer PGD (WiderNet Surrogate)
• Boundary Search Diagnostic
V9 Gain: 2.08% Transfer ASR
55.2% fewer boundary transitions

D. Margin Proof & Sweeps

• Unconstrained DeepFool Budget
(L2 = 4.15 vs 2.04 -> 2.03x distance;
97.3% flips exceed eps=0.05)
• Epsilon Sweep (0.01 -> 0.10)
• PGD Scaling (20 -> 200 steps)

Setup: Baseline Model vs. Our Proposed Defense (V9)